BUSINESS REQUIREMENTS DOCUMENT

BeOne POC

Feature Summaries
Prepared by Kolosal AI

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BUSINESS REQUIREMENTS DOCUMENT

01 - Onboarding

Prepared by Kolosal AI

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1. Frontend

1.1 Overview

The Onboarding feature provides a personalized, role-aware landing experience for users immediately after they log in. It ensures that the system recognizes the user's identity and permissions before granting access to the main dashboard, providing a secure and tailored entry point into the application.

1.2 Step-by-Step Flow

1. Authentication Loading: Upon logging in, the system displays a "Signing in…" screen with a loading spinner. This confirms to the user that the system is securely verifying their credentials and gathering their profile information.
2. Data Retrieval: The system automatically fetches the user's profile details (such as name and role) and their current task status (such as pending drafts or items needing review).
3. Personalized Welcome: Once the data is loaded, the user is directed to a welcome screen. This screen displays a personalized greeting, the user's profile avatar, and clear badges indicating their assigned roles within the system.
4. Task Overview: The dashboard provides a summary of the user's current workload, highlighting items that require their attention.
5. Navigation: The user can click a "Continue to Dashboard" button to proceed to the main management area. This button includes a loading state to ensure a smooth transition and prevent accidental double-clicks.

1.3 Key Concepts

- Role-Awareness: The interface automatically adjusts to show specific information and permissions based on the user's assigned role, ensuring they only see what is relevant to their job function.
- Global State: A centralized way the application remembers who the user is and what their current tasks are, so this information is instantly available across different screens.
- Task Summary: A snapshot of the user's active work, such as documents in progress or items waiting for approval, allowing for quick status checks.
- Loading State: A visual indicator (spinner) that manages user expectations during the brief period when the system is communicating with the server to fetch user data.

1.4 Business Value

- Efficiency: By providing a personalized task summary immediately upon login, users can quickly identify their priorities without searching through the entire system.

- Security & Compliance: The role-aware design ensures that users are correctly identified and categorized, maintaining strict adherence to organizational permissions and access controls.
- User Experience: The smooth transition from login to the dashboard, combined with clear visual feedback (like loading spinners and role badges), creates a professional and reliable experience that reduces user frustration.

2. Logic

2.1 Overview

The Onboarding feature provides a personalized landing page for users immediately after they log in. By automatically identifying who the user is and what their specific role is, the system tailors the dashboard to show relevant information. This ensures that every user sees a welcome message, their profile details, and a summary of their current task status as soon as they access the application.

2.2 Step-by-Step Flow

1. Identity Verification: When a user accesses the dashboard, the system checks their secure login session to confirm their identity. If the session is invalid or missing, access is denied.
2. Profile Retrieval: Once verified, the system pulls the user's core profile information, including their name, avatar, assigned role, and regional settings (such as time zone and language).
3. Task Summary Compilation: The system gathers real-time data regarding the user's current workload, specifically counting items in progress, items awaiting review, rejected items, and items ready for publication.
4. Dashboard Initialization: The system combines the profile and task data into a single package and sends it to the dashboard, which then displays the personalized welcome screen and task metrics.

2.3 Key Concepts

- Authenticated Session: A secure digital "handshake" that confirms the user is who they say they are, allowing the system to safely retrieve their private data.
- Role-Awareness: The system's ability to recognize the user's specific job function (e.g., admin vs. member) to ensure they see the correct interface and information.
- Task Summary: A high-level snapshot of a user's workflow, providing immediate visibility into what needs their attention without requiring them to navigate through multiple screens.
- Regional Settings: User-specific preferences, such as time zone and language, that ensure the application displays dates, times, and text in a way that is familiar and accurate for the individual user.

2.4 Business Value

This feature improves user efficiency by eliminating the need to manually search for status updates upon login. By presenting a "ready-to-work" dashboard, the system reduces cognitive load and helps users immediately identify their priorities. Furthermore, by automatically applying role-based settings, the application ensures that users are only interacting with the tools and data relevant to their specific responsibilities, which enhances both security and user experience.

3. Data

3.1 Overview

The "Onboarding" feature provides a personalized welcome experience for users as soon as they log into the system. Instead of loading a generic page, the application immediately identifies who the user is, what their role is, and what tasks require their attention. This ensures that the dashboard is ready to use the moment the user arrives, without requiring them to click through multiple screens to see their current workload.

3.2 Step-by-Step Flow

1. Authentication Check: The system verifies the user's identity through their active session or login token. If the user is not logged in, access is denied.
2. Profile Retrieval: Once verified, the system pulls the user's core profile information, including their name, contact details, assigned roles, and regional preferences (like time zone and language).
3. Task Aggregation: The system scans the content modules to count the user's active work items, such as drafts, items waiting for review, rejected work, and items ready to be published.
4. Dashboard Initialization: The system combines the profile data and the task counts into a single package and sends it to the user's screen, instantly populating the dashboard with their specific information.

3.3 Key Concepts

- Authenticated Session: The secure connection that confirms a user's identity, allowing the system to provide personalized content.
- Role-Awareness: The system recognizes the user's specific job function (e.g., Editor or Reviewer) to ensure they see the correct tools and information relevant to their responsibilities.
- Task Summary: A real-time snapshot of a user's workload, displayed as badges or counts on the dashboard to help them prioritize their day.
- Bootstrap: The process of loading all necessary user information at once during the initial login, which makes the application feel faster and more responsive.

3.4 Business Value

- Improved User Productivity: By surfacing pending tasks immediately upon login, users can identify their priorities without searching through the system.
- Reduced Friction: Consolidating profile and task data into a single step eliminates the need for multiple page loads, creating a smoother and more professional user experience.
- Personalized Access Control: By leveraging role data, the system ensures that users only interact with the features and content appropriate for their specific job level.

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02 - ICF Dashboard

Prepared by Kolosal AI

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1. Frontend

1.1 Overview

The Consent IQ Dashboard serves as the central command center for managing Informed Consent Forms (ICFs). It provides a high-level summary of your consent activity and a structured list of all protocols, allowing users to easily search, sort, and access specific documents for review or editing.

1.2 Step-by-Step Flow

1. Dashboard Initialization: Upon loading, the system automatically retrieves and displays summary statistics, including the total number of consents, those currently in review, and those already approved.
2. Navigation: Users can initiate the creation of a new consent form by clicking the "Create New Consent" button in the header.
3. Searching: Users can type into the search bar to instantly filter the list of protocols by name or ICF details. Clearing the search box restores the full list.
4. Sorting: Users can click on any column header in the table to sort the data. Clicking a header toggles the order between ascending and descending; a visual indicator (²/¼) shows the current sort status.
5. Managing Documents: Each protocol row provides action buttons:

* View: Opens a preview of the document.

* Open: Navigates to the editing interface for drafts or documents currently in review.

* Archived Items: Protocols marked as "Archived" are read-only and do not display action buttons.

1.3 Key Concepts

- Consent IQ Dashboard: The main interface for tracking and managing the lifecycle of consent forms.
- Protocol Table: A dynamic list displaying key details such as protocol IDs, versions, current status, and last modification dates.
- Status Indicators: Visual badges that categorize each document (e.g., Draft, In Review, Approved) to help users quickly identify the progress of a protocol.
- Dynamic Search: A real-time filtering tool that updates the table view as you type to help you locate specific protocols or ICFs quickly.
- Sortable Columns: A feature that allows users to organize the table data by clicking headers, making it easier to find the most recently modified or specific versions of documents.

1.4 Business Value

This dashboard streamlines the administrative workflow by consolidating all consent-related tasks into one view. By providing instant access to status metrics and intuitive search and sort capabilities, it reduces the time spent manually tracking documents. This ensures that teams can efficiently monitor the progress of consent forms, quickly identify items requiring attention, and maintain better oversight of the entire protocol lifecycle.

2. Logic

2.1 Overview

The ICF Dashboard serves as the central command center for managing consent forms. It provides users with a high-level summary of their workload while offering powerful tools to locate and organize specific protocols and their associated documents. The dashboard is designed to be responsive, allowing users to quickly filter through data and monitor the progress of consent forms through their lifecycle.

2.2 Step-by-Step Flow

1. Dashboard Initialization: When the page loads, the system automatically retrieves summary statistics, including the total number of protocols and a breakdown of those currently "In Review" versus "Approved."
2. Data Retrieval: The system fetches the full list of protocols, pulling in relevant details such as version numbers, status, and associated file information.
3. Sorting: Users can click on column headers to organize the list. The system dynamically updates the view to sort data in either ascending or descending order based on the selected category.
4. Searching: As a user types into the search bar, the system instantly filters the list. It looks for matches across both protocol names and consent form names, narrowing down the results in real-time to help the user find exactly what they need.

2.3 Key Concepts

- Protocol: The primary record representing a clinical study, which contains specific versions, status updates, and associated documentation.
- ICF (Informed Consent Form): The specific document or file linked to a protocol that is being tracked and managed within the system.
- Dashboard Statistics: A high-level snapshot provided at the top of the screen that gives immediate visibility into the volume and status of all active work.
- Dynamic Sorting: The ability to reorder the list by clicking headers, allowing users to prioritize information (e.g., by status or date).
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Instant Search: A responsive search feature that filters results as the user types, eliminating the need to press "enter" or wait for a page refresh.

2.4 Business Value

- Improved Efficiency: By providing instant search and sorting, users spend less time hunting for documents and more time managing them.
- Enhanced Visibility: The dashboard statistics provide immediate insight into the current workload, helping teams identify bottlenecks (such as a high volume of forms "In Review") at a glance.
- Centralized Management: Consolidating protocol data and file information into one view ensures that users have a single source of truth, reducing the risk of errors or version confusion.

3. Data

3.1 Overview

The ICF Dashboard serves as the central command center for managing Informed Consent Forms (ICFs) across various research protocols. It provides users with a high-level summary of their current workload and a detailed, searchable list of all consent forms, allowing for efficient tracking and organization of documentation.

3.2 Step-by-Step Flow

1. Dashboard Initialization: When you first open the dashboard, the system automatically calculates and displays summary statistics, including the total number of protocols and a breakdown of how many are currently "In Review" versus "Approved."
2. Data Retrieval: The system fetches the full list of protocols, displaying key details such as the protocol name, version, associated ICF name, current status, and the date of the last modification.
3. Searching: You can enter a specific protocol name or ICF name into the search bar. The system will filter the list in real-time to show only the items that match your criteria.
4. Sorting: You can organize the list by clicking on column headers (such as status or last modified date) to sort the information in either ascending or descending order, helping you prioritize your work.

3.3 Key Concepts

- Protocol: A formal research study plan that serves as the primary container for all associated consent documentation.
- ICF (Informed Consent Form): The specific document used to obtain participant consent, which is tracked and managed within the context of its parent protocol.
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Dashboard Stats: A quick-reference summary that provides immediate visibility into the volume and progress of your consent forms.

– Filtering & Sorting: Tools that allow you to narrow down large lists of data and arrange them in a way that makes the most sense for your current task.

3.4 Business Value

This feature eliminates the need to manually track consent forms across disparate systems. By providing a unified view, it reduces administrative burden, minimizes the risk of missing critical updates, and ensures that team members can quickly identify which protocols require immediate attention based on their current status.

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03 - ICF Creation - Setup

Prepared by Kolosal AI

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1. Frontend

1.1 Overview

The ICF Creation Setup page is a three-column wizard designed to prepare the necessary information for the AI to generate an Informed Consent Form (ICF). Users walk through a structured process to identify the clinical study, provide supporting documentation, and select the appropriate document format.

1.2 Step-by-Step Flow

1. Select Protocol: The user begins by choosing a clinical protocol from a list. A search bar is available to filter protocols by name or ID. Once a protocol is selected, it is highlighted to confirm the choice.
2. Attach Reference Documents: The user provides context for the AI by attaching an "Investigation Brochure" and a "Lab Manual." These can be uploaded directly from the user's device or fetched from the document management system.
3. Select Template: The user chooses a document template to define the structure of the final ICF. A "standard" template is selected by default, and the system highlights any templates recommended for the chosen protocol.
4. Validation: The system ensures a protocol is selected before allowing document uploads. If any errors occur (such as a failed upload or missing selection), an error banner will appear to guide the user.

1.3 Key Concepts

- Protocol: The specific clinical study or trial for which the ICF is being generated.
- Reference Documents: Supporting files (like brochures or manuals) that provide the AI with the necessary background information to draft accurate content.
- Template: A pre-defined document structure that ensures the generated ICF meets specific formatting and regulatory requirements.
- Client-side Filtering: A feature that allows the user to instantly narrow down the list of protocols as they type, without needing to reload the page.

1.4 Business Value

This feature streamlines the ICF creation process by centralizing all required inputs into a single, intuitive interface. By requiring users to link specific reference documents to a protocol and select a standardized template, the system ensures that the AI has high-quality, relevant data to work with. This reduces manual effort, minimizes the risk of using incorrect study information, and ensures consistency across all generated consent forms.

2. Logic

2.1 Overview

The ICF (Informed Consent Form) Creation wizard is a three-step interface designed to help users prepare the necessary information for AI-assisted document generation. By guiding the user through selecting a clinical protocol, attaching relevant reference materials, and choosing a document template, the system ensures that the AI has the correct context to produce accurate and compliant documentation.

2.2 Step-by-Step Flow

1. Protocol Selection: Upon opening the wizard, the system displays a list of available clinical protocols. Users can browse the full list or use the search bar to find a specific protocol by name or ID.
2. Reference Document Attachment: Once a protocol is selected, the user provides the necessary reference materials (such as an Investigator's Brochure or Lab Manual). Users have two options for each document type:

* Upload: Directly upload a file from their local device.

* Fetch from DMS: Retrieve existing documents already stored in the Document Management System (DMS) associated with the selected protocol.

3. Template Selection: The user selects a document template from a provided catalog. The system highlights recommended templates to assist the user in making the best choice for their specific needs.
4. Validation: Throughout the process, the system ensures that a protocol is selected before allowing document retrieval, providing clear feedback if a step is missed or if an upload fails.

2.3 Key Concepts

– Protocol: The specific clinical study or trial framework that serves as the foundation for the document being generated.
– Reference Documents: Supporting files (like manuals or brochures) that provide the AI with the necessary background information to generate accurate content.
– DMS (Document Management System): A centralized repository where existing project documents are stored and can be retrieved for use in the wizard.
– Template: A pre-defined document structure or "blueprint" that dictates the format and style of the final generated ICF.

2.4 Business Value

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Efficiency: By centralizing protocol selection, document retrieval, and template choice into a single wizard, the time required to prepare for document generation is significantly reduced.

- Accuracy: Providing the AI with specific, verified reference documents ensures that the generated ICF is contextually relevant and aligned with the correct clinical protocol.
- Consistency: Using standardized templates ensures that all generated ICFs maintain a professional, uniform structure, reducing the risk of manual formatting errors and improving compliance.

3. Data

3.1 Overview

The ICF (Informed Consent Form) Creation Setup feature is a guided, three-step wizard designed to streamline the preparation of new consent documents. By centralizing the selection of clinical protocols, reference materials, and document templates, the system ensures that users have all the necessary components ready before triggering the AI-driven generation process.

3.2 Step-by-Step Flow

1. Protocol Selection: Upon opening the wizard, the system displays a catalog of available clinical protocols. Users can browse the full list or use a search function to quickly find a specific protocol by its name or ID.
2. Reference Document Attachment: Once a protocol is selected, the user can view or upload relevant reference documents. The system retrieves existing documents associated with that specific protocol and allows for the addition of new files, ensuring all source material is linked correctly.
3. Template Selection: The user chooses an appropriate document template from a curated list. The system highlights "recommended" templates to help users select the most suitable format for their specific needs.
4. Launch Generation: With the protocol, reference documents, and template confirmed, the setup is complete, and the user can proceed to launch the AI generation process.

3.3 Key Concepts

- Protocol: The foundational clinical study framework that dictates the requirements for the consent form.
- Reference Document: Supporting files (such as study summaries or safety information) that provide the necessary context for the AI to generate an accurate consent form.
- Template: A pre-defined document structure that serves as the starting point for the AI, ensuring consistency and compliance across different consent forms.
- Wizard: A user-friendly interface that breaks a complex task into manageable, sequential steps to prevent errors and improve efficiency.

3.4 Business Value

- Efficiency: Reduces the time spent manually gathering study information and formatting documents by automating the setup phase.
- Consistency: By using standardized templates and linking specific reference documents to protocols, the organization ensures that all generated consent forms meet internal quality and compliance standards.
- User Experience: The guided wizard approach minimizes the learning curve for new users and reduces the likelihood of missing critical information during the document preparation stage.

BUSINESS REQUIREMENTS DOCUMENT

05 - ICF Creation - Edit and Refine

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1. Frontend

1.1 Overview

The "Edit and Refine" feature provides a user-friendly workspace for reviewing and polishing AI-generated Informed Consent Forms (ICFs). It allows users to view the document structure, manually edit specific sections, and trigger AI-assisted refinements to ensure the content is accurate and professional.

1.2 Step-by-Step Flow

1. Document Loading: Upon opening the feature, the system retrieves the ICF document and displays it in a structured layout.
2. Navigation: Users can browse the document using a Table of Contents sidebar, which lists all sections for easy access.
3. Section Review: The main canvas displays the document as a series of cards. Each card shows the section title, a badge indicating if it was AI-generated, and an "Edit with AI" button.
4. Selection: When a user clicks "Edit with AI" on a specific section, that content is loaded into a dedicated side panel.
5. Editing: In the side panel, the user can manually modify the text. The system includes built-in validation to ensure that sections are not saved as empty.
6. Refinement: Once the user is satisfied with their manual edits or requests an AI update, they can save the changes to update the document.

1.3 Key Concepts

- Section-Based Editing: The document is broken down into manageable parts, allowing users to focus on one specific area at a time rather than the entire document at once.
- AI-Assisted Refinement: A dedicated tool that helps users improve or rewrite specific sections of the document using AI capabilities.
- Live Validation: A safety feature that prevents users from accidentally saving empty or incomplete sections, ensuring document integrity.
- Visual Status Indicators: Clear labels (such as "AI Generated") and loading spinners that keep the user informed about the document's state and any ongoing AI processing.

1.4 Business Value

This feature streamlines the document creation process by combining the efficiency of AI generation with the precision of human oversight. By providing a dedicated, intuitive interface for editing, it reduces the time required

to finalize complex legal and medical documents while ensuring that the final output meets specific quality and compliance standards.

2. Logic

2.1 Overview

The "Edit and Refine" feature provides a dedicated workspace for reviewing and polishing AI-generated Informed Consent Forms (ICFs). When you open a document, the system automatically retrieves the latest version of the content and its associated metadata. This ensures that you are always working with the most up-to-date information, allowing you to seamlessly transition between viewing the document structure and making manual or AI-assisted refinements.

2.2 Step-by-Step Flow

1. Document Initialization: When you navigate to the edit view, the system automatically requests the specific document by its unique identifier.
2. Data Retrieval: The system fetches the full document package, which includes the document name, current status, version history, and the actual text content.
3. Loading State: While the system gathers this information, a loading indicator is displayed to ensure you know the document is being prepared for your review.
4. Display: Once the data is retrieved, the document is rendered in the center panel, organizing the content into distinct sections that are ready for your review, manual edits, or AI-assisted updates.

2.3 Key Concepts

- Document Metadata: Information about the document, such as its name, version number, and current status, which helps you track the progress of the ICF.
- Content Retrieval: The process of pulling the latest version of the document text from the system so that your edits are applied to the most recent draft.
- Section-Based Editing: A layout approach where the document is broken down into manageable parts, allowing you to focus on specific areas for refinement without needing to manage the entire document at once.

2.4 Business Value

This feature streamlines the document review process by providing a reliable, "single source of truth" for your ICFs. By automating the retrieval and organization of document sections, it reduces the administrative burden on users,

minimizes the risk of working on outdated drafts, and provides a clear, intuitive interface for finalizing high-quality consent forms efficiently.

3. Data

3.1 Overview

The "Edit and Refine" feature provides a user-friendly interface for managing Informed Consent Form (ICF) documents. It allows users to review AI-generated content, make manual adjustments, and leverage AI assistance to rewrite specific sections of the document. This ensures that the final document is accurate, compliant, and tailored to the specific needs of the study.

3.2 Step-by-Step Flow

1. Document Retrieval: When you open a document, the system fetches the full document details, including its current content, version information, and status, to display it in the editor.
2. Manual Editing: You can directly modify the text within the document. The system validates your changes to ensure the document remains in a valid state and saves your updates automatically.
3. AI-Assisted Refinement: If you need help with a specific section, you can select that section and provide instructions (a prompt) to the AI.
4. Content Update: The AI generates a refined version of that specific section based on your instructions, which is then returned to the editor for your review and final approval.
5. Persistence: Every time a change is made—whether manual or AI-assisted—the system updates the document record and logs the time of the last modification.

3.3 Key Concepts

- Manual Edit: The ability for a user to type directly into the document to correct or update information.
- AI-Assisted Rewrite: A feature that uses artificial intelligence to rephrase or improve specific parts of the document based on user-provided guidance.
- Document State: The current version of the ICF, including its text, metadata, and the timestamp of the most recent change.
- Section Management: The capability to target specific parts of the document for editing or AI refinement without affecting the entire document.

3.4 Business Value

This feature significantly reduces the administrative burden of drafting ICFs by combining the speed of AI generation with the precision of human oversight. By allowing for both manual control and AI-powered suggestions,

the tool ensures high-quality, compliant documentation while drastically shortening the time required to finalize complex consent forms.

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07 - ICF Creation - Preview

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1. Frontend

1.1 Overview

The ICF Preview feature provides a final "read-only" view of the Informed Consent Form (ICF) before it is officially published. It allows users to review the document layout and content in a simulated PDF format, ensuring everything is accurate before finalizing the document.

1.2 Step-by-Step Flow

1. Loading: When the user opens the preview page, the system automatically retrieves the document details and all associated content sections.
2. Review: The document is displayed in a simulated PDF view, allowing the user to read the full text and verify the structure.
3. Adjustment: Users can use the toolbar to zoom in or out to better inspect specific parts of the document.
4. Navigation: If the user identifies any errors, they can click the "Back to Review" button to return to the previous editing stage.
5. Publishing: Once the user is satisfied, they click the "Publish" button. Upon success, the button is replaced by a confirmation badge, indicating the document is now finalized.

1.3 Key Concepts

- Simulated PDF View: A visual representation of the final document that mimics how the ICF will appear once generated, allowing for a final quality check.
- Document Toolbar: A control area that displays the document title and page count, while providing zoom tools for better readability.
- Publishing: The final action that locks the document content and officially issues the ICF.
- Back to Review: A safety mechanism that allows users to exit the preview and return to the editing phase if they spot mistakes.

1.4 Business Value

This feature acts as a final quality assurance gate. By providing a clear, accurate preview of the document, it reduces the risk of errors in legal or clinical documentation. It empowers users to confidently finalize their work, knowing exactly what the end result will look like before it is officially published.

2. Logic

2.1 Overview

The ICF Preview and Publication feature allows users to review a document in a simulated PDF format before finalizing it. This ensures that the document content and layout are correct before it is officially released. Once the user is satisfied with the preview, they can trigger the publication process to lock the document and mark it as complete.

2.2 Step-by-Step Flow

1. Loading the Preview: When you navigate to the preview screen, the system automatically gathers the document's details (such as the title and status) and all its individual content sections.
2. Reviewing the Document: The gathered information is displayed in a simulated PDF view, allowing you to verify the document's appearance and content.
3. Initiating Publication: Once you have confirmed the document is ready and all necessary approvals are in place, you click the "Publish" button.
4. Finalizing: The system processes the request to lock the document. Upon success, the interface updates to display a "ICF Published Successfully" badge, confirming the document is now officially published with a recorded timestamp.

2.3 Key Concepts

- Metadata: The high-level information about your document, such as its title, current status, and creation date.
- Sections: The individual blocks of content that make up the full document, which are assembled together for the preview.
- Publication: The final step in the document lifecycle that locks the content, preventing further edits and marking the document as officially released.
- Approval Gates: Pre-defined checks that must be satisfied before the system allows you to publish the document, ensuring compliance and quality control.

2.4 Business Value

This feature provides a "what-you-see-is-what-you-get" experience, reducing errors by allowing stakeholders to perform a final quality check before a document is made official. By automating the publication process and providing clear status updates, the system ensures that document records are accurate, timestamped, and protected from unauthorized changes after they have been finalized.

BUSINESS REQUIREMENTS DOCUMENT

04 - ICF Creation - Generate

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1. Frontend

1.1 Overview

The ICF Generation feature provides a transparent, real-time view of the document creation process. When a user initiates the generation of an Informed Consent Form (ICF), the system transitions to a dedicated dashboard that tracks progress, estimates the remaining time, and highlights specific milestones as they are completed. Once the document is fully generated, the system automatically redirects the user to the editor to review the final output.

1.2 Step-by-Step Flow

1. Initiation: The user triggers the generation process, and the system begins monitoring the status of the request.
2. Real-Time Tracking: The dashboard displays a progress bar and a countdown timer (ETA) that updates automatically as the system works.
3. Milestone Visualization: The system moves through six distinct stages:

* Parsing protocol data
* Loading reference documents
* Applying regulatory templates
* Generating consent sections
* Running compliance checks
* Finalizing document

4. Status Updates: Each stage is visually marked as "In progress" (with an animated spinner), "Done" (with a green checkmark), or "Pending" (for future steps).
5. Completion: Once the final stage is reached and the progress hits 100%, the system automatically navigates the user to the document editor to view the completed ICF.

1.3 Key Concepts

- Progress Tracking: A visual indicator showing the percentage of completion to keep the user informed of the system's activity.
- ETA Display: A dynamic countdown timer that estimates how much time is left until the document is ready, helping manage user expectations.
- Sequential Checkpoints: A list of specific tasks the AI performs, providing transparency into the "behind-the-scenes" work required to build the document.
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Auto-Navigation: A seamless transition that moves the user from the generation screen to the editor as soon as the file is ready, removing the need for manual interaction.

1.4 Business Value

This feature significantly improves the user experience by eliminating the "black box" effect during document generation. By providing clear, step-by-step feedback and an estimated time of arrival, users feel more confident in the system's reliability. Furthermore, the automatic transition to the editor reduces friction, allowing users to move immediately from generation to review without unnecessary clicks or waiting.

2. Logic

2.1 Overview

The ICF Generation feature provides a transparent, real-time view of the AI document creation process. Instead of leaving the user waiting on a static screen, the system displays a progress bar and a list of six specific checkpoints. As the AI works, the interface updates dynamically to show exactly which stage of the generation is currently active, ensuring the user is kept informed until the document is ready for review.

2.2 Step-by-Step Flow

1. Initialization: Once the generation process begins, the system establishes a live connection to monitor the AI's progress.
2. Real-time Updates: As the AI completes tasks, the system receives live updates. The progress bar advances, and the specific checkpoint currently being processed is highlighted on the screen.
3. Completion: Once the AI finishes all six steps, the system marks the process as complete.
4. Auto-Navigation: After a brief pause to allow the user to see the final status, the system automatically redirects the user to the editor screen, where they can review and refine the newly created document.
5. Error Handling: If the connection is interrupted or the generation fails, the system stops the process and displays an error message, allowing the user to address the issue.

2.3 Key Concepts

- Live Progress Tracking: A visual indicator that keeps the user updated on the status of the AI generation in real-time.
- Checkpoints: A set of six distinct milestones that break down the complex generation process into understandable stages.
- Auto-Navigation: A seamless transition that moves the user from the generation screen to the editing workspace as soon as the document is ready.

– Connection Monitoring: A background process that ensures the user is alerted immediately if the generation encounters a technical issue or connection loss.

2.4 Business Value

This feature improves the user experience by removing the uncertainty often associated with AI-driven tasks. By providing clear, step-by-step visibility, users feel more confident in the system's reliability. Furthermore, the automatic transition to the editor reduces friction, allowing users to move directly from document creation to the refinement phase without unnecessary manual clicks.

BUSINESS REQUIREMENTS DOCUMENT

06 - ICF Creation - Review

Prepared by Kolosal AI

Version 1.0 **Date** October 13, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

The ICF Review page is the final stage in the document creation process. It provides a centralized workspace where reviewers can examine the draft document, track its history, and provide feedback. The interface is designed to be intuitive, separating the document content from the administrative tools needed to finalize the ICF.

1.2 Step-by-Step Flow

1. Document Loading: Upon entering the page, the system retrieves the full document content and the complete history of actions taken on that document.
2. Document Visualization: The document is displayed in the center of the screen, organized into distinct sections. Each section is presented as a card, clearly identifying whether the content was generated by AI.
3. Navigation: A table of contents on the left allows users to jump quickly to specific sections of the document.
4. Feedback Loop: Reviewers use the "Comments" tab on the right panel to provide specific feedback or suggestions for changes.
5. History Tracking: Reviewers can switch to the "Audit Trail" tab to view a chronological record of all activities, including who made edits, when comments were posted, and the overall progression of the document.
6. Finalization: Once the review is complete, the reviewer uses the bottom navigation bar to move the document forward in the workflow.

1.3 Key Concepts

- Section Cards: A modular way of displaying document content that highlights AI-generated text, making it easy for human reviewers to identify areas that may require closer scrutiny.
- Audit Trail: A transparent, time-stamped log of every significant action taken on the document, ensuring full accountability and traceability for compliance purposes.
- Comment Panel: A dedicated space for collaborative feedback, allowing reviewers to communicate directly within the context of the document.
- Progress Stepper: A visual indicator at the top of the page that shows the current stage of the document within the broader creation workflow.

1.4 Business Value

This feature ensures high-quality, compliant ICF documents by facilitating a structured review process. By providing clear visibility into AI-generated content and maintaining a comprehensive audit trail, the system reduces the risk of

errors and ensures that all feedback is captured and addressed. This transparency supports regulatory compliance and streamlines the transition from draft to final document.

2. Logic

2.1 Overview

The ICF Review feature provides a dedicated workspace for stakeholders to examine draft Informed Consent Forms (ICFs). This interface allows reviewers to view the current content of a document alongside a comprehensive history of all actions taken on that document. By combining the document's current state with a clear audit trail, the system ensures that reviewers have the necessary context to provide feedback and track the document's progress through the approval lifecycle.

2.2 Step-by-Step Flow

1. Document Loading: When a user opens the Review Page, the system automatically retrieves the full document details, including its current content, status, and version information.
2. Content Display: The system renders the document in the center panel, organizing the information into clear sections so the reviewer can easily read and evaluate the draft.
3. Audit Trail Retrieval: Simultaneously, the system fetches the document's activity history, which tracks every significant action taken on the file.
4. History Presentation: The audit information is displayed in a chronological feed, showing the most recent actions at the top. This allows the reviewer to see who performed specific actions and when they occurred.

2.3 Key Concepts

- Document Metadata: The essential details of the ICF, such as its name, version, current status, and timestamps, which help identify the specific draft being reviewed.
- Audit Trail: A chronological record of all interactions with the document, providing transparency and accountability by logging who performed an action and when.
- Review Interface: The visual layout of the page that separates the document content from its history, allowing for a focused review process.

2.4 Business Value

- Improved Accuracy: By providing a clear view of the document and its history, reviewers can make more informed decisions when providing feedback.
- Compliance and Traceability: The audit trail ensures that every change or review action is documented, which is critical for maintaining regulatory standards and internal accountability.

- Efficiency: Centralizing the document content and its history in one view reduces the time spent searching for context, allowing for a faster and more streamlined review cycle.

3. Data

3.1 Overview

The ICF Review feature allows users to access a draft version of an Informed Consent Form (ICF) to evaluate its content and address feedback. This functionality provides a centralized view where a reviewer can see the current state of the document alongside a complete history of actions taken on it, ensuring transparency and accountability throughout the review process.

3.2 Step-by-Step Flow

1. Document Retrieval: When a user opens the review screen, the system fetches the full details of the selected ICF, including its current status, content, and version information.
2. Displaying Content: The system populates the main workspace with the document's text and metadata, allowing the reviewer to read the draft and identify areas requiring updates.
3. Audit Trail Loading: Simultaneously, the system retrieves a chronological log of all previous activities related to that specific document.
4. Review and Feedback: The reviewer examines the document and the audit history (which shows who made changes and when) to understand the context of the current draft and incorporate necessary feedback.

3.3 Key Concepts

- Document State: The complete set of information regarding an ICF, including its current version, status, and the actual text content being reviewed.
- Audit Trail: A chronological record of all actions performed on a document, providing a clear timeline of edits and reviews to help users track the document's evolution.
- Metadata: Supporting information attached to the document, such as creation dates and version identifiers, which helps reviewers verify they are looking at the correct draft.

3.4 Business Value

This feature streamlines the quality control process for Informed Consent Forms. By providing a clear view of the document content alongside a transparent history of changes, it reduces the risk of errors, ensures that reviewer feedback is properly addressed, and maintains a reliable record of the document's lifecycle for compliance and audit purposes.

BUSINESS REQUIREMENTS DOCUMENT

Global Navigation Header

Prepared by Kolosal AI

Version 1.0 **Date** October 13, 2026 **Status** Draft **Classification** Internal

1. components

1.1 Overview

The Global Navigation Header is a persistent, fixed-position bar that appears at the top of every page within the Consent IQ platform. It serves two primary purposes: establishing brand identity and providing users with quick access to their personal account information and notification status.

1.2 Step-by-Step Flow

1. Initialization: When the application loads, the system automatically requests the current user's information, including their name, assigned role, and any pending notifications.
2. Brand Display: The left side of the header displays the platform branding, including the "Consent IQ" title, a descriptive subtitle, and the "Powered by Agilisium" attribution.
3. Notification Check: The system checks for unread tasks. If any exist, a notification bell icon appears with a badge indicating the specific number of pending items.
4. User Identity Display: The right side of the header displays the user's profile information, including an avatar based on their initials, their full name, and their current role. If the system cannot retrieve specific user data, it defaults to generic placeholders to ensure the interface remains clean and functional.
5. Interaction: A dropdown menu icon (chevron) is provided next to the user profile to allow for future expansion of account-related actions.

1.3 Key Concepts

- User Context: The collection of personal details (name, role, and notification count) that the system retrieves to personalize the header for each logged-in user.
- Fallback Logic: A safety mechanism that ensures the header displays default text (such as "User Name" or "ICF Author") if the system is unable to retrieve live data, preventing empty or broken-looking fields.
- Persistent Header: A design element that remains fixed at the top of the screen as the user navigates through different parts of the application, ensuring consistent access to navigation and status updates.
- Notification Badge: A visual indicator attached to the notification icon that dynamically updates to show the number of unread items, helping users stay informed about pending tasks.

1.4 Business Value

This feature provides a professional and consistent user experience across the entire Consent IQ platform. By centralizing branding and user-specific information, it reinforces the platform's identity while allowing users to immediately gauge their workload through the notification badge. The use of automated data loading and fallback

defaults ensures that the interface remains reliable and user-friendly, even if there are temporary issues with data retrieval.

2. Data

2.1 Overview

The Global Navigation Header is a consistent top-bar component designed to provide users with immediate access to their identity and status within the platform. It ensures that regardless of which page a user is on, they can see their name, professional role, and a summary of their current workload. The system is designed to be resilient, providing sensible default information if a user's profile data is temporarily unavailable.

2.2 Step-by-Step Flow

1. The system identifies the currently logged-in user based on their active session.
2. The system retrieves the user's profile details, including their name, primary and secondary roles, and the number of tasks currently assigned to them.
3. If the system successfully finds the user, it displays their specific information in the header.
4. If the system cannot find the user (or if data is missing), it automatically displays a default profile: "User Name" with a role of "ICF Author" and zero tasks.
5. The header displays this information alongside a dropdown menu icon, allowing the user to interact with their profile settings.

2.3 Key Concepts

- User Context: The collection of personal details (name, roles, and task volume) that the system uses to personalize the navigation experience.
- Resilience (Fallback Logic): A safety mechanism that ensures the navigation header remains functional and professional even if the system encounters an error or cannot retrieve specific user data.
- Task Summary: A quick-reference indicator that shows the user how many items currently require their attention.

2.4 Business Value

- Improved User Experience: By providing a consistent header across the platform, users always know who is logged in and can easily access their profile.
- Operational Efficiency: Displaying task counts directly in the header helps users stay informed about their workload without needing to navigate to a separate dashboard.
-

System Reliability: The fallback mechanism ensures that the interface remains clean and usable even during unexpected data issues, preventing broken layouts or confusing empty states for the end user.

BUSINESS REQUIREMENTS DOCUMENT

Progress Stepper

Prepared by Kolosal AI

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1. Components

1.1 Overview

The Progress Stepper is a visual navigation tool displayed at the top of the creation screens in the BeOne platform. It provides users with a clear roadmap of the five-stage creation process, helping them understand their current progress and allowing them to jump back to previous stages if needed.

1.2 Step-by-Step Flow

1. Display: The system renders a horizontal bar showing five distinct stages: Setup, Generate, Edit & Refine, Review, and Preview.
2. Status Identification: Each step is automatically assigned a status based on the user's progress:

* Done: Steps already completed (marked with a check icon).

* Active: The current step the user is working on (highlighted for visibility).

* Future: Steps that have not yet been reached (grayed out).

3. Navigation: Users can click on any "Done" or "Active" step to return to that screen, provided the following safety rules are met:

* The "Generate" step is a temporary process and cannot be clicked.

* Advanced steps (Edit, Review, and Preview) can only be accessed if a document has been successfully created.

1.3 Key Concepts

- Step Status: A visual indicator that tells the user whether a stage is finished, currently in progress, or upcoming.
- Transient Step: A specific stage in the process that acts as a bridge or background task; because it is not a static page, users cannot navigate back to it directly.
- Conditional Navigation: A security feature that ensures users can only access later stages of the process once the necessary data (the document) has been generated.

1.4 Business Value

The Progress Stepper improves the user experience by reducing confusion during complex workflows. By clearly showing the path ahead and allowing users to easily revisit previous steps, it minimizes errors and helps users feel in control of the creation process. This modular design also ensures that users cannot skip essential steps or access advanced features before the system is ready to support them.

BUSINESS REQUIREMENTS DOCUMENT

Wrapper Shell

Prepared by Kolosal AI

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1. components

1.1 Overview

The Wrapper Shell is a consistent structural container designed for the BeOne POC creation flow. It acts as a "frame" that surrounds every page in the five-step process, ensuring that users always have access to the same navigation tools, status indicators, and project metadata regardless of which step they are currently viewing.

1.2 Step-by-Step Flow

1. Initialization: When a user navigates to a page within the creation flow, the system identifies the current step and any available project details (such as the Protocol ID, ICF name, and version).
2. Navigation Display: The shell renders a breadcrumb bar at the top, allowing the user to easily return to the main Dashboard or identify their current location within the "Consent Creation" process.
3. Metadata Injection: The system checks which project details are available. If data exists, it displays the Protocol ID, ICF name, and version as visual "pills" or labels to provide immediate context.
4. Progress Tracking: A step indicator is displayed to show the user exactly where they are in the five-step sequence.
5. Content Rendering: The specific page content for the current step is injected into the center of the shell, ensuring the surrounding navigation and metadata remain fixed and visible.

1.3 Key Concepts

- Wrapper Shell: A standardized layout template that ensures a consistent look and feel across different parts of the application.
- Breadcrumbs: A navigation aid that shows the user's path, allowing them to jump back to previous levels like the Dashboard.
- Conditional Metadata: Information (like Protocol ID or version) that only appears on the screen if that specific data has been provided, keeping the interface clean.
- Step Indicator: A visual progress bar that helps the user understand how much of the five-step creation process they have completed.

1.4 Business Value

- Improved User Experience: By providing a consistent layout, users don't have to re-orient themselves when moving between steps, reducing confusion and errors.
- Context Awareness: Displaying key project identifiers (like Protocol ID and version) directly in the header ensures that users always know exactly which project they are working on without needing to navigate away.
- Efficiency: The standardized progress indicator helps users track their work, leading to a more predictable and efficient completion of the consent creation process.

BUSINESS REQUIREMENTS DOCUMENT

Table of Contents

Prepared by Kolosal AI

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1. components

1.1 Overview

The Table of Contents (TOC) feature provides a navigation panel on the left side of the document viewer. It acts as a roadmap for the document, allowing users to quickly jump to specific sections. The TOC is dynamic, meaning it automatically updates to highlight the section currently being viewed, ensuring users always know their location within the document.

1.2 Step-by-Step Flow

1. Document Parsing: The system scans the document sections and identifies their structure.
2. Hierarchy Formatting: The system checks section titles for specific numbering patterns (e.g., "3.1"). If a pattern is found, the item is treated as a sub-section and indented in the list; otherwise, it is treated as a main section.
3. Display: The TOC is rendered in the left panel, showing the list of sections with their corresponding numbers and titles.
4. Navigation: When a user clicks a section title in the TOC, the main document view smoothly scrolls to that specific section.
5. Active State Tracking: As the user scrolls through the document, the system monitors which section is currently visible on the screen and automatically highlights that corresponding item in the TOC.

1.3 Key Concepts

- Dynamic Hierarchy: The TOC automatically distinguishes between main sections and sub-sections based on the title format, creating a clean, indented visual structure.
- Active State: A visual indicator that keeps the TOC in sync with the user's current position in the document, providing helpful context during long reads.
- Smooth Navigation: A "click-to-scroll" interaction that allows users to bypass manual scrolling and jump directly to the content they need.

1.4 Business Value

This feature significantly improves the user experience by reducing the time spent searching for information within long documents. By providing a clear, interactive navigation aid, users can maintain focus and efficiency, ensuring they can easily reference or jump between critical sections of the document without losing their place.

2. LOGIC

2.1 Overview

The Table of Contents (TOC) feature provides a navigation panel on the left side of your document workspace. It acts as a dynamic outline, allowing you to view, organize, and quickly jump to specific parts of your document. The system automatically keeps this list updated as you add new sections or subsections, ensuring you always have a clear view of your document's structure.

2.2 Step-by-Step Flow

1. Loading the Document: When you open a document, the system automatically retrieves all existing sections and displays them in the left-hand panel, ordered by their position in the document.
2. Adding Content: You can click the "Add Section" or "Add Subsection" buttons to create new parts of your document.
3. Automatic Updates: Once a new section is created, the system instantly adds it to your Table of Contents list.
4. Navigation: The system automatically selects the newly created section and scrolls your main view directly to that area, so you can begin editing immediately.
5. Visual Cues: If a section was generated or assisted by AI, a small badge will appear next to its title in the Table of Contents, helping you identify AI-authored content at a glance.

2.3 Key Concepts

- Section: A distinct block of content within your document that appears as an entry in the Table of Contents.
- Table of Contents (TOC): The sidebar navigation panel that lists all document sections in their correct order.
- Active Section: The specific part of the document you are currently viewing or editing, highlighted in the sidebar for easy reference.
- AI Label: A visual indicator (badge) that marks content created by the AI, providing transparency regarding the source of the text.

2.4 Business Value

- Improved Navigation: Users can jump to specific document areas instantly, saving time spent scrolling through long documents.
- Enhanced Organization: By providing a clear, hierarchical view of sections and subsections, the feature helps users maintain a logical structure throughout the document creation process.
- Transparency: The AI badge ensures users are always aware of which parts of the document were generated by AI, supporting better content management and review workflows.
- Seamless Workflow: The automatic scrolling and selection features reduce manual effort, allowing users to focus on writing rather than managing the document interface.

3. DATA

3.1 Overview

The Table of Contents (TOC) feature provides a dynamic navigation panel on the left side of the document editor. It allows users to view the structure of their document, including all sections and subsections, in the correct order. This feature ensures that as users add or organize content, the navigation panel stays synchronized, making it easy to jump to specific parts of the document.

3.2 Step-by-Step Flow

1. Initial Load: When a document is opened, the system automatically retrieves the full list of sections associated with that document.
2. Ordering: The system organizes these sections based on their assigned position, ensuring the Table of Contents matches the document's layout.
3. Display: The retrieved list is rendered in the left-hand panel, providing a clear overview of the document structure.
4. Dynamic Updates: When a user adds a new section or subsection, the system calculates its correct position in the hierarchy and saves it.
5. Refresh: Immediately after a new section is created, the Table of Contents updates to include the new entry, ensuring the navigation panel is always current.

3.3 Key Concepts

- Section: A distinct block of content within a document that can be tracked and navigated to via the Table of Contents.
- Hierarchy: The relationship between sections and subsections, which determines how the document is structured and displayed in the navigation panel.
- Positioning: A system-managed order that ensures sections appear in the Table of Contents exactly as they are intended to appear in the document.
- Synchronization: The process of automatically updating the Table of Contents whenever the document structure changes, such as adding a new section.

3.4 Business Value

- Improved Navigation: Users can quickly jump to specific sections of long or complex documents, saving time and reducing manual scrolling.
-

Enhanced Document Clarity: By providing a visual outline, the Table of Contents helps users maintain a clear understanding of the document's structure and flow.

- Seamless Editing Experience: Because the Table of Contents updates automatically when new sections are added, users can focus on content creation without needing to manually manage or refresh their navigation tools.

BUSINESS REQUIREMENTS DOCUMENT

Bottom Bar - Setup

Prepared by Kolosal AI

Version 1.0 **Date** October 13, 2026 **Status** Draft **Classification** Internal

1. COMPONENTS

1.1 Overview

The Bottom Action Bar on the "Setup" page provides a consistent navigation and action area for users. It allows users to easily return to the main dashboard or initiate the ICF generation process once all necessary setup requirements have been met.

1.2 Step-by-Step Flow

1. Validation: The system continuously monitors the setup form. The "Generate ICF" button remains inactive until a protocol is selected, at least one reference is attached, and a template is chosen.
2. Navigation: Users can click the "Back to Dashboard" button at any time to exit the setup process and return to the main dashboard.
3. Initiation: Once the form is valid, the "Generate ICF" button becomes active. Clicking this button triggers the generation process.
4. Transition: Upon a successful start to the generation process, the system automatically redirects the user to the progress visualization page to track the status of their document creation.
5. Error Handling: If the generation process fails, the system will display a notification to the user, allowing them to attempt the action again.

1.3 Key Concepts

- Conditional Activation: A safety feature that ensures the "Generate" action is only available when the required information is complete, preventing errors.
- Navigation Controls: Dedicated buttons that provide clear paths for users to either move forward with their task or exit to the dashboard.
- Progress Tracking: A seamless transition from the setup phase to a dedicated view where users can monitor the real-time status of their document generation.

1.4 Business Value

This feature streamlines the user experience by providing clear, context-aware actions. By enforcing form validation before allowing the generation process to start, it reduces the likelihood of errors and ensures that the system always has the necessary data to produce high-quality ICF drafts. The intuitive layout helps users navigate the setup process efficiently, minimizing friction and improving overall productivity.

2. logic

2.1 Overview

The "Generate ICF" feature, located in the bottom action bar of the Setup page, allows users to initiate the creation of an Informed Consent Form (ICF) draft. By selecting a protocol and a template, the system automatically extracts and structures the necessary information from your documents to prepare the draft for further processing.

2.2 Step-by-Step Flow

1. Initiation: Once you have selected a protocol, attached the required references, and chosen a template, the "Generate ICF" button becomes active.
2. Request Submission: Clicking the button sends your request to the system to begin the generation process.
3. Job Creation: The system creates a new "Generation Job" to track the progress of your request and immediately confirms that the process has started.
4. Document Parsing: In the background, the system retrieves the relevant document, extracts the text, and organizes it into a structured format.
5. Downstream Processing: Once the document is successfully parsed, the structured data is automatically passed to the next stage of the pipeline to finalize the ICF draft.

2.3 Key Concepts

- Generation Job: A tracking record created by the system to manage your request from start to finish, ensuring you know the current status of your document generation.
- Protocol: The specific clinical study or research framework that provides the source material for the ICF.
- Template: A pre-defined layout or structure that dictates how the final ICF document should be formatted.
- Parsing: The automated process of reading a raw document and identifying its individual sections and data points so the system can use that information to build the new form.

2.4 Business Value

This feature streamlines the document creation process by removing the need for manual data entry. By automating the extraction and structuring of information from existing protocols, the system reduces the risk of human error, ensures consistency across all generated forms, and significantly accelerates the time it takes to prepare an ICF draft for review.

3. Data

3.1 Overview

The "Setup" page bottom action bar includes a feature that allows users to initiate the generation of an Informed Consent Form (ICF). When a user triggers this action, the system begins a background process to create the document, allowing the user to continue their work without waiting for the file to finish generating.

3.2 Step-by-Step Flow

1. Initiation: The user clicks the generate action on the Setup page.
2. Validation: The system checks the provided information to ensure it is complete and accurate.
3. Job Creation: The system creates a new "pending" task in the background to handle the document generation.
4. Confirmation: The system immediately sends a confirmation back to the user, providing a unique job ID that identifies this specific generation request.
5. Background Processing: The system works in the background to finalize the document, keeping the user interface responsive.

3.3 Key Concepts

- ICF Generation: The automated process of creating an Informed Consent Form based on the specific protocol and template selected by the user.
- Background Processing: A method where the system performs complex tasks "behind the scenes," ensuring the user does not have to wait on a loading screen.
- Job ID: A unique reference number assigned to each generation request, which allows the system to track the progress of that specific document.
- Pending Status: The initial state of a request, indicating that the system has received the instruction and is preparing to process it.

3.4 Business Value

This feature improves user productivity by removing the need to wait for document generation to complete before moving to the next task. By handling the process in the background, the application remains fast and responsive, providing a smoother experience for users managing complex protocol setups.

BUSINESS REQUIREMENTS DOCUMENT

Bottom Bar - Edit

Prepared by Kolosal AI

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1. components

1.1 Overview

The "Edit and Refine" page features a persistent navigation bar at the bottom of the screen. This bar provides users with clear, accessible options to either return to the previous configuration stage or finalize their work by submitting the document for formal review.

1.2 Step-by-Step Flow

1. Navigation Options: The user is presented with two primary actions: "Back to Setup" and "Submit for Review."
2. Returning to Setup: If the user selects "Back to Setup," the system immediately navigates them back to the initial configuration page for the current protocol.
3. Submitting for Review: If the user selects "Submit for Review," the system initiates a request to update the document's status from "Draft" to "In Review."
4. Processing State: While the submission is being processed, the "Submit for Review" button is temporarily disabled to prevent duplicate requests.
5. Completion: Upon a successful submission, the system automatically redirects the user to the review page for that specific document.

1.3 Key Concepts

- Persistent Action Bar: A fixed navigation area at the bottom of the screen that ensures essential controls are always available to the user, regardless of how much content is on the page.
- Document Status Transition: The automated process of moving a document from a "Draft" state to an "In Review" state, signaling that the user has finished their edits.
- Submission Guard: A safety feature that disables the submit button during processing to ensure data integrity and prevent accidental multiple submissions.

1.4 Business Value

This feature streamlines the user workflow by providing a clear path to completion. By placing navigation and submission controls in a consistent, easy-to-reach location, we reduce user effort and minimize the risk of errors. The automated status transition ensures that the document lifecycle is managed accurately, allowing for a seamless handoff from the editing phase to the review phase.

2. logic

2.1 Overview

The "Submit for Review" feature allows users to finalize their work on the "Edit and Refine" page. By clicking this button, the user signals that the document is ready for the next stage of the workflow, moving it from a draft state into an official review process.

2.2 Step-by-Step Flow

1. Initiation: The user clicks the "Submit for Review" button located in the bottom action bar.
2. Verification: The system checks to ensure the document exists and is valid for submission.
3. Status Update: The system automatically updates the document's status from "Draft" to "In Review" and records the time of the update.
4. Confirmation: The system confirms the update and prepares the document for the next phase of the workflow.

2.3 Key Concepts

- Document Status: A label that tracks where a document is in its lifecycle (e.g., moving from a working draft to a formal review).
- In Review: The specific status assigned to a document once the user has finished editing and submitted it for approval.
- Workflow Transition: The automated process of moving a document from one stage to another, ensuring the system always reflects the current state of the work.

2.4 Business Value

This feature streamlines the transition between editing and approval, reducing manual administrative work. By automating the status change, it ensures that stakeholders are immediately notified that a document is ready for review, which helps maintain project momentum and ensures that documents move through the pipeline efficiently without unnecessary delays.

3. Data

3.1 Overview

The "Edit and Refine" page features a bottom action bar designed to help users manage their documents efficiently. This interface provides two primary functions: saving ongoing work as a draft and submitting a completed document for formal approval. The system ensures that documents move through a controlled workflow, preventing accidental submissions and maintaining clear status tracking.

3.2 Step-by-Step Flow

1.

Saving Progress: When a user makes changes to a document, they can save their work. The system updates the document content and metadata while keeping the document in its current state.

2. Initiating Submission: When the user is ready to finalize their work, they trigger the submission action from the bottom bar.
3. Status Verification: The system automatically checks the document's current status. It will only proceed if the document is currently marked as a "Draft."
4. Finalizing Submission: If the document is a valid draft, the system updates its status to "In Review," signaling that it is ready for the next stage of the process. If the document is not in a draft state, the system will prevent the submission to avoid errors.

3.3 Key Concepts

- Draft: The initial state of a document where the user is actively editing content.
- In Review: The status assigned to a document once it has been submitted, indicating it is no longer editable by the author and is awaiting further action.
- Status Transition: The controlled movement of a document from one stage (Draft) to another (In Review) to ensure data integrity.
- Metadata: Additional descriptive information attached to the document that is saved alongside the content during updates.

3.4 Business Value

- Data Integrity: By enforcing status checks, the system prevents users from accidentally submitting documents that are not ready or are already in the review process.
- Workflow Efficiency: The bottom action bar provides a clear, consistent user experience, allowing authors to easily toggle between saving their progress and moving their work forward.
- Audit Readiness: Every update and status change is tracked with a timestamp, ensuring there is a clear record of when a document was modified or submitted for review.

BUSINESS REQUIREMENTS DOCUMENT

Bottom Bar - Generate

Prepared by Kolosal AI

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1. COMPONENTS

1.1 Overview

The "Generate" page features a fixed navigation bar at the bottom of the screen designed to give users control over their workflow. This bar provides two clear options that allow users to either return to the initial configuration phase or bypass the current generation process entirely to move directly to manual editing.

1.2 Step-by-Step Flow

1. The user is presented with two action buttons at the bottom of the "Generate" page.
2. If the user selects "Back to Setup," the system navigates them away from the current page and returns them to the initial configuration screen.
3. If the user selects "Skip," the system bypasses the generation process and takes the user directly to the manual editing screen for their specific project.

1.3 Key Concepts

- Back to Setup: A navigation option that allows users to restart or adjust their initial project settings.
- Skip: A navigation option that allows users to bypass the automated generation step and proceed immediately to manual refinements.
- Job ID: A unique identifier that ensures the system remembers which project the user is working on when they choose to skip ahead.

1.4 Business Value

This feature improves user experience by providing flexibility and control. By allowing users to easily return to previous steps or skip unnecessary generation tasks, the interface reduces frustration and helps users reach their desired editing environment more efficiently. It ensures that the workflow remains non-linear, accommodating different user needs and project requirements.

BUSINESS REQUIREMENTS DOCUMENT

Bottom Bar - Review

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1. Components

1.1 Overview

The Review Bottom Bar is a persistent navigation tool located at the bottom of the document review screen. Its primary purpose is to provide users with a clear path to either return to the editing phase or finalize their work by submitting the document for formal approval. The bar is designed to be context-aware, providing immediate feedback regarding any outstanding tasks before a user commits to the final submission.

1.2 Step-by-Step Flow

1. Reviewing the Document: The user is on the Review page, where they can see the document and any associated feedback or comments.
2. Navigation Option: The user can select the "Back to Edit" button if they decide further changes are required, which immediately returns them to the editing interface.
3. Submission Readiness: The system checks for any unresolved comments. If comments remain open, the "Submit for Approval" button updates its label to display the specific number of open items, alerting the user to pending tasks.
4. Final Submission: When the user clicks "Submit for Approval," the system initiates the formal submission process.
5. Processing and Feedback: While the document is being processed, the button is temporarily disabled to prevent duplicate submissions. Once the process completes, the user receives a notification confirming whether the submission was successful or if an error occurred.

1.3 Key Concepts

- Sticky Navigation: A fixed bar at the bottom of the screen that remains accessible regardless of how far the user has scrolled through the document.
- Contextual Labeling: The submission button dynamically updates its text based on the number of open comments, ensuring the user is aware of incomplete work before finalizing.
- Submission Guardrails: A built-in safety feature that disables the submission button during the processing phase to prevent accidental multiple clicks or system errors.

1.4 Business Value

- Improved Data Quality: By highlighting the number of open comments directly on the submission button, the system encourages users to address feedback before finalizing, reducing the likelihood of submitting incomplete or inaccurate documents.
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Enhanced User Experience: Providing a persistent, easy-to-reach navigation bar reduces friction, allowing users to move seamlessly between editing and reviewing without losing their place.

- Operational Efficiency: Automating the submission process ensures that documents are routed for approval consistently and reliably, while immediate feedback on success or failure keeps the user informed of the document's status in real-time.

2. Logic

2.1 Overview

The "Submit for Approval" feature provides a clear, guided path for users to finalize their work on the Review page. It ensures that documents are only moved forward once all necessary feedback has been addressed, preventing incomplete documents from progressing to the next stage of the workflow.

2.2 Step-by-Step Flow

1. Validation Check: When you click the "Submit for Approval" button, the system automatically checks if there are any unresolved comments on the document.
2. Conditional Blocking: If open comments exist, the system prevents the submission and notifies you. If no open comments remain, the system proceeds to the next step.
3. Status Update: The system updates the document status to "Approved."
4. Automatic Navigation: Upon a successful update, you are automatically redirected to the Preview page to view the final version of your document.

2.3 Key Concepts

- Gate Condition: A safety check that ensures specific requirements (like resolving all comments) are met before a document can move to the next stage.
- Dynamic Labeling: The button text updates based on the document's status; if comments are still open, the button displays the count of those comments to keep you informed.
- Workflow Automation: The seamless transition from the Review page to the Preview page, which reduces manual steps and ensures you are always looking at the most relevant screen.

2.4 Business Value

This feature improves data quality and process efficiency by enforcing a "quality gate." By requiring all comments to be resolved before submission, the system reduces the risk of errors and ensures that only finalized, reviewed content moves forward. This creates a smoother, more reliable experience for users and maintains the integrity of the document approval process.

BUSINESS REQUIREMENTS DOCUMENT

AI Document Editor

Prepared by Kolosal AI

Version 1.0 **Date** October 13, 2026 **Status** Draft **Classification** Internal

1. COMPONENTS

1.1 Overview

The AI Document Editor is a feature within the BeOne platform that allows users to refine and polish document content using artificial intelligence. It provides a dedicated workspace where users can manually edit text or use AI-driven prompts to automatically rewrite, simplify, or summarize sections of a document.

1.2 Step-by-Step Flow

1. Review and Edit: The user selects a document section, which is displayed in an editable text area. The user can make manual changes directly to the text before involving the AI.
2. Provide Instructions: The user can either type a custom instruction into the prompt box or select a "Quick Prompt" (such as simplifying language for patient readability) to automatically populate the instruction field.
3. Trigger AI: Once the user clicks the "Ask AI" button, the system first saves any manual edits made to the text to ensure no work is lost.
4. Processing: The system sends the current text and the user's instructions to the AI service. The button updates to show it is "Asking..." to indicate that the request is in progress.
5. Update: Once the AI completes the rewrite, the document section is instantly updated with the new content, and the interface refreshes to reflect these changes.

1.3 Key Concepts

- Manual Refinement: The ability for users to adjust text manually, ensuring they maintain control over the final document output.
- Quick Prompts: Pre-defined, one-click instructions that help users quickly apply common formatting or readability improvements without needing to type out a request.
- State Synchronization: A background process that ensures the document content remains consistent across the editor and the rest of the application, preventing data discrepancies.
- Instruction-Based Editing: A natural language approach where the AI acts on specific user requests to modify the tone, length, or clarity of the text.

1.4 Business Value

- Increased Efficiency: Reduces the time spent on manual drafting and editing by automating repetitive writing tasks.
- Improved Quality: Ensures document content is clear, professional, and tailored to the intended audience (e.g., simplifying complex medical or technical language for patients).

- Seamless Workflow: By integrating the AI directly into the editing panel and automatically saving manual work, the tool provides a smooth, uninterrupted experience that minimizes the risk of data loss.

2. Logic

2.1 Overview

The AI Document Editor feature allows users to refine specific sections of a document using natural language instructions. By selecting a section and providing a prompt (such as "simplify this language" or "check for compliance"), the system leverages AI to rewrite the content while maintaining the context of the original document. This provides a seamless way to polish text without leaving the editing interface.

2.2 Step-by-Step Flow

1. Input: The user selects a specific document section and enters a natural language instruction into the AI editor.
2. Context Gathering: The system retrieves the current content and title of the selected section to ensure the AI understands what it is editing.
3. AI Processing: The system sends the section content and the user's instructions to an AI service, which acts as a professional editor to generate the requested refinements.
4. Update: The system receives the refined text from the AI and saves it as a new version of that document section.
5. Display: The updated content is returned to the user interface, replacing the previous version so the user can immediately review the changes.

2.3 Key Concepts

- Natural Language Instruction: A plain-English request from the user that tells the AI how to modify or improve the text.
- Contextual Awareness: The ability of the system to provide the AI with the existing document content, ensuring the edits are relevant and accurate to the specific section.
- Version Control: Each time the AI refines a section, the system creates a new version, allowing for tracking and potential recovery of previous content.
- Refinement Status: A marker applied to the section once the AI has successfully processed the request, indicating that the content has been updated by the AI.

2.4 Business Value

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Increased Efficiency: Users can quickly rewrite, summarize, or adjust the tone of document sections without manual drafting.

- Improved Quality: Professional-grade editing assistance helps ensure documents are clear, consistent, and compliant with organizational standards.
- Seamless Workflow: By integrating AI directly into the document editor, users stay within their current task flow, reducing the need to switch between different tools or applications.

3. Data

3.1 Overview

The AI Document Editor feature allows users to refine specific sections of a document using artificial intelligence. When a user provides a prompt to improve or rewrite a section, the system automatically processes the request, updates the content, and maintains a clear audit trail of the changes. This ensures that users can easily iterate on their documents while keeping track of which parts have been modified by AI.

3.2 Step-by-Step Flow

1. Request Initiation: The user submits a request to refine a specific document section by providing a prompt (instructions on how to change the text) and the current content of that section.
2. AI Processing: The system sends the user's prompt and the existing text to the AI engine, which generates the refined version of the content.
3. Version Tracking: The system retrieves the current version of the section and increments the version number to ensure a history of changes is maintained.
4. Update and Labeling: The system updates the section with the new content, tags it as "AI Edited," and sets the status to "AI Generated."
5. Persistence: The updated section is saved to the database, and the new version is returned to the user for review.

3.3 Key Concepts

- Refinement Prompt: The specific instructions provided by the user to guide the AI in rewriting or improving the document text.
- Version Control: A system that automatically increments a counter every time a section is updated, allowing users to track the evolution of their document.
- Provenance Tagging: A labeling system that identifies whether a section was modified by AI or manually, providing transparency regarding the source of the content.

- Status Tracking: A workflow state that indicates the current condition of a section (e.g., "AI Generated"), helping users manage the document review process.

3.4 Business Value

- Increased Efficiency: Users can quickly rewrite or polish complex document sections without manual drafting, significantly reducing the time spent on editing.
- Improved Accuracy and Consistency: By using AI to refine text, users can ensure that document sections remain clear, professional, and aligned with organizational standards.
- Auditability: The automatic versioning and labeling features provide a clear history of changes, which is essential for maintaining document integrity and compliance in professional environments.

BUSINESS REQUIREMENTS DOCUMENT

AI Openrouter SDK

Prepared by Kolosal AI

Version 1.0 **Date** October 13, 2026 **Status** Draft **Classification** Internal

1. FRONTEND

1.1 Overview

The AI Document Generation feature allows users to create professional documents automatically by combining specific templates with custom data. By leveraging advanced AI models through the OpenRouter platform, the system can take your input, process it against a chosen template, and produce a structured, ready-to-use document. The system also includes a built-in parser that breaks down complex files into manageable sections, ensuring the AI can accurately interpret and revise your content.

1.2 Step-by-Step Flow

1. Input Collection: You provide the necessary details, including the template ID, the source file, the specific AI model you wish to use, and any configuration data required for the document.
2. Validation: The system automatically checks your inputs in real-time to ensure all required fields are present and correctly formatted before processing begins.
3. Document Parsing: Once submitted, the system extracts the raw text from your source file and organizes it into logical sections, preparing the content for AI interaction.
4. AI Generation: The system sends your instructions and the structured document data to the AI model. The AI processes this information to generate the final document content.
5. Completion: Upon success, the system saves the generated document and provides you with the final output, including the document ID and the completed text. If an error occurs, the system will notify you with a clear message.

1.3 Key Concepts

- Template ID: A unique identifier that tells the system which document layout or structure to use for your request.
- Source Filename: The original file containing the raw information that needs to be processed or transformed.
- AI Model: The specific "brain" or engine (such as GPT-4) selected to perform the document generation and editing tasks.
- Configuration Values: A set of specific data points or instructions provided in a structured format that the AI uses to fill in the blanks within your template.
- Document Parsing: The process of reading a file and breaking it down into smaller, logical parts so the AI can understand and edit specific sections accurately.

1.4 Business Value

This feature significantly reduces the manual effort required to draft and format complex documents. By automating the assembly process, it ensures consistency across all generated files, minimizes human error, and allows your team to produce high-quality documentation in a fraction of the time. The ability to choose different AI models provides flexibility, allowing you to balance performance and cost based on the complexity of the document being created.

2. LOGIC

2.1 Overview

The AI OpenRouter SDK feature enables the system to automatically process, analyze, and summarize documents using advanced AI models. By connecting to the OpenRouter platform, the system can ingest PDF files, extract their content, and generate structured summaries or revisions based on specific user instructions. This allows for automated document handling, reducing the need for manual review and data entry.

2.2 Step-by-Step Flow

1. File Preparation: The system selects the target document and converts it into a digital format (base64 encoding) that the AI can read and process.
2. AI Connection: The system establishes a secure connection to the OpenRouter platform using a pre-configured API key.
3. Content Submission: The document and specific processing instructions are sent to a chosen AI model (such as Gemini or Claude).
4. AI Analysis: The AI model analyzes the document content and generates a summary or structured output based on the provided instructions.
5. Data Storage: The system receives the AI-generated content, saves the summary to the database, and breaks the output into logical sections for easy retrieval and future use.

2.3 Key Concepts

- OpenRouter: A platform that provides access to various high-performance AI models through a single, unified connection.
- Base64 Encoding: A method of converting a file (like a PDF) into a text-based format so it can be easily transmitted to the AI service.
- AI Model: The "brain" behind the feature (e.g., Gemini 1.5 Pro) that performs the actual reading, summarizing, and structuring of the document.
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Structured Output: The process of taking raw AI-generated text and organizing it into specific, labeled sections (like "Summary" or "Key Findings") so the system can store and display it effectively.

2.4 Business Value

- Efficiency: Automates the time-consuming task of reading and summarizing long documents.
- Consistency: Ensures that document analysis is performed using standardized instructions, reducing human error.
- Scalability: Allows the business to process large volumes of documents quickly without increasing manual workload.
- Flexibility: By using OpenRouter, the system can easily switch between different AI models to find the best balance of cost, speed, and accuracy for specific document types.

3. DATA

3.1 Overview

The AI OpenRouter SDK feature enables our system to automatically process and generate documents by leveraging advanced AI models. By connecting to the OpenRouter platform, the system can ingest PDF files, analyze their content, and produce structured, high-quality document outputs that are saved directly into our database for future use.

3.2 Step-by-Step Flow

1. File Preparation: The system takes a source PDF document and converts it into a digital format (base64) that the AI can read and process.
2. AI Analysis: The system sends the document and specific instructions to an AI model via the OpenRouter interface.
3. Content Generation: The AI analyzes the document and generates the requested content, which is returned to our system.
4. Record Creation: The system creates a primary record in the database to store metadata about the document, such as the template used, the model selected, and the total page count.
5. Structured Storage: The system breaks the generated content into individual, organized sections (e.g., titles and body text) and saves them into the database, ensuring the information is easy to retrieve and manage later.

3.3 Key Concepts

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OpenRouter: A service that provides access to various powerful AI models through a single, unified connection.

- Base64: A standard method for encoding files into text, allowing documents like PDFs to be sent securely to AI services.
- Document Generation: The process of using AI to create or fill out document templates based on provided source material.
- Structured Data: Organizing the AI's output into specific sections (like titles and content) rather than one long block of text, which makes the information more useful for our applications.

3.4 Business Value

- Efficiency: Automates the time-consuming task of manually extracting and formatting information from complex documents.
- Consistency: Ensures that generated documents follow a standardized structure and quality, regardless of the source material.
- Scalability: By using a flexible AI integration, we can easily swap or upgrade models as technology improves without needing to rebuild our entire document processing workflow.
- Accessibility: Storing content in a structured database format allows other parts of our software to easily search, display, or report on the generated document data.

BUSINESS REQUIREMENTS DOCUMENT

Edit Actions

Prepared by Kolosal AI

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1. COMPONENTS

1.1 Overview

The Edit Actions Bar is a control panel located on the "Edit and Refine" page. It provides users with the necessary tools to manage specific sections of a document while maintaining clear visibility of the document's identity and version. This feature ensures that users can easily modify, remove, or review content within a structured and safe environment.

1.2 Step-by-Step Flow

1. Context Display: The bar displays the document name and version, along with the title of the section currently selected by the user.
2. Action Selection: The user chooses one of three primary actions:

* Preview Draft: Opens a full-screen modal that displays the entire document as a PDF, allowing for a final review of the layout.
* Edit: Switches the selected section into an interactive text-editing mode. The user can then save their changes (which updates the system) or cancel to discard them.
* Delete: Triggers a confirmation window to prevent accidental data loss. If the user confirms, the section is permanently removed from the document.

3. System Feedback: After performing an action like saving or deleting, the system provides a notification to confirm the success or failure of the operation and refreshes the document view accordingly.

1.3 Key Concepts

- Document Context: The header information that keeps the user oriented by showing the document name and version.
- Section Editing: An interactive mode that allows users to modify specific content blocks and choose whether to commit those changes to the system.
- Preview Modal: A read-only, full-screen view of the document that helps users visualize how their changes appear in the final document format.
- Delete Confirmation: A safety mechanism that requires an extra step before a section is removed, ensuring that important content is not deleted by mistake.

1.4 Business Value

- Efficiency: By providing direct access to edit and preview functions, users can refine document content without navigating away from their current workspace.

- Data Integrity: The inclusion of a confirmation step for deletions and a "cancel" option for edits protects against accidental data loss or unintended changes.
- User Confidence: Displaying the document version and providing a full-document preview ensures that users are always working on the correct iteration and can verify their work in the final format before finalizing.

2. logic

2.1 Overview

The Edit Actions feature provides a streamlined way for users to manage document content directly within the "Edit and Refine" page. It allows users to view document details, modify specific sections of text, and remove sections that are no longer needed. The system ensures that all changes are tracked for auditing purposes and that the document structure remains organized and consistent after any edits.

2.2 Step-by-Step Flow

1. Document Loading: When the page opens, the system retrieves the full document, including its metadata, status, and all individual sections, to display them in the center panel.
2. Unique Identification: When a new document is created, the system automatically generates a unique identifier (e.g., PROTO-2026-782-ICF) to ensure every document has a distinct, traceable name.
3. Editing Content: When a user edits a section, the system saves the new text. Simultaneously, it creates an entry in the activity log to record that a manual edit occurred, ensuring a clear audit trail.
4. Deleting Sections: If a user chooses to delete a section, the system removes it permanently. It then automatically re-sequences the remaining sections to ensure the document maintains a logical, unbroken order.

2.3 Key Concepts

- Audit Trail: A background process that automatically records manual changes to document content, providing accountability and a history of modifications.
- Section Re-sequencing: An automated cleanup process that runs after a deletion to ensure that the remaining parts of the document are correctly numbered and ordered.
- Unique Identifier: A standardized, system-generated naming convention that prevents duplicate document names and ensures each file is easily searchable and distinct.
- Persistence: The process of saving changes (like text edits or deletions) to the system so they remain available after the user leaves the page.

2.4 Business Value

- Data Integrity: By automatically re-sequencing sections after a deletion, the system prevents structural errors and ensures the document remains professional and readable.
- Compliance and Accountability: The automatic logging of manual edits provides a reliable audit trail, which is essential for tracking changes in sensitive or regulated documentation.
- User Efficiency: The intuitive edit bar allows users to perform common tasks—like updating text or removing sections—without needing to navigate away from the document, significantly speeding up the refinement process.

3. data

3.1 Overview

The Edit Action Bar is a core component of the "Edit and Refine" page, designed to give users direct control over document content. It provides a streamlined interface for modifying or removing specific parts of a document, ensuring that all changes are accurately captured and saved to the system.

3.2 Step-by-Step Flow

1. Loading the Document: When a user opens the "Edit and Refine" page, the system automatically retrieves the full document, including its metadata, status, and current content, to display it in the center panel.
2. Editing Content: When a user makes changes to a specific section, the system captures these updates and saves them to the server. Every manual edit is logged by the system for auditing and tracking purposes.
3. Deleting Content: If a user chooses to remove a section, the system triggers a confirmation prompt to prevent accidental deletion. Once confirmed, the section is permanently removed, and the document structure is updated to reflect the change.
4. Processing State: Throughout these actions, the system maintains a "processing" state to ensure that the user interface remains responsive and prevents duplicate requests while an action is being completed.

3.3 Key Concepts

- Document Metadata: Information about the document, such as its name, version, and status, which helps the system manage and track the document throughout its lifecycle.
- Section-Level Control: The ability to interact with individual parts of a document independently, allowing for precise editing and management without affecting the entire document at once.
- Audit Logging: A background process that records manual edits, ensuring there is a clear history of changes made to the document for compliance and transparency.
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Confirmation Prompt: A safety mechanism that requires the user to verify their intent before performing irreversible actions, such as deleting a section.

3.4 Business Value

This feature enhances user productivity by providing a reliable and intuitive way to refine documents. By separating the document into manageable sections and providing clear, logged actions for editing and deletion, the system reduces the risk of errors, ensures data integrity, and maintains a transparent audit trail for all document modifications.

BUSINESS REQUIREMENTS DOCUMENT

Comment Actions

Prepared by Kolosal AI

Version 1.0 **Date** October 13, 2026 **Status** Draft **Classification** Internal

1. components

1.1 Overview

The Comments & Approval feature provides a centralized workspace on the document review page to manage feedback and finalize document approval. It allows team members to discuss specific sections, track the status of feedback, and ensures that all necessary reviews and resolutions are completed before a document can be officially submitted for approval.

1.2 Step-by-Step Flow

1. Adding Feedback: Users can add comments using the pinned composer at the bottom of the panel. When typing, they can associate the comment with a specific document section.
2. Managing Conversations: Comments are displayed in a threaded list. Each comment shows the author, timestamp, and the relevant section. Users can reply to existing comments to keep discussions organized.
3. Updating Status: Each comment has a status toggle (Open, Resolved, or Closed). Team members can update these statuses directly within the comment card to reflect the current state of the discussion.
4. Reviewer Actions: The Approval Gate tracks whether all assigned reviewers have completed their required actions. Reviewers can update their status via the interface.
5. Final Submission: The "Submit for Approval" button monitors the document's readiness. It remains disabled until two conditions are met: all comments are marked as "Resolved" or "Closed," and all reviewers have completed their actions. Once these criteria are satisfied, the button becomes active, allowing the user to submit the document.

1.3 Key Concepts

- Comment Threading: A structured way to view original comments and their subsequent replies, keeping related feedback grouped together.
- Status Toggle: A control that allows users to move a comment from "Open" to "Resolved" or "Closed," which directly impacts the document's readiness for approval.
- Approval Gate: A validation check that prevents a document from being submitted until all feedback is addressed and all reviewers have provided their input.
- Section Association: The ability to link a comment to a specific part of the document, ensuring context is preserved for the reader.

1.4 Business Value

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Quality Assurance: By requiring all comments to be resolved before submission, the system prevents documents from moving forward with outstanding issues or unanswered questions.

- Accountability: The Approval Gate ensures that every assigned reviewer has performed their duty, creating a clear audit trail for the document lifecycle.
- Efficiency: Centralizing communication and status tracking in one panel reduces the need for external emails or meetings, allowing teams to resolve feedback in real-time within the document context.

2. Logic

2.1 Overview

The Comment Actions feature provides a dedicated space on the document review page for team members to communicate, track feedback, and manage the approval process. It allows users to view threaded conversations, post new comments or replies, and update their individual review status, ensuring that all feedback is addressed before a document is submitted for final approval.

2.2 Step-by-Step Flow

1. Loading Comments: When a user opens the document, the system automatically retrieves all existing comments and replies, displaying them in a threaded list within the right-hand panel.
2. Adding Feedback: Users can add new comments or reply to existing ones using the pinned composer at the bottom of the panel. Once submitted, the new entry is saved and immediately appears in the thread.
3. Managing Review Status: Within the approval area, reviewers can click an "Act" button to confirm their participation. The system updates their status to "acted" in real-time.
4. Approval Validation: The system continuously monitors two conditions: that all comments are marked as "Resolved" or "Closed," and that every reviewer has performed their "Act" action.
5. Submission: Once all comments are resolved and all reviewers have acted, the "Submit for Approval" button becomes active, allowing the user to finalize the document review.

2.3 Key Concepts

- Threaded Comments: A structured view where replies are nested under their original parent comment, keeping conversations organized and easy to follow.
- Reviewer Status: A tracking mechanism that records whether a specific reviewer has completed their required actions on the document.
- Approval Gate: A safety check that prevents a document from being submitted until all feedback is addressed and all reviewers have signed off.

- Real-time Updates: The ability for the comment thread to refresh instantly when a new comment is added, without requiring the user to manually reload the page.

2.4 Business Value

This feature streamlines the review cycle by centralizing communication and accountability. By enforcing an "Approval Gate," it ensures that no document is submitted prematurely, reducing the risk of overlooked feedback or incomplete reviews. The real-time nature of the comment thread fosters faster collaboration, allowing teams to resolve issues efficiently and move projects forward with confidence.

3. Data

3.1 Overview

The Comment Actions feature provides a dedicated space on the document review page for users to collaborate. It allows reviewers to post comments, reply to existing discussions, and track the resolution status of specific feedback. Additionally, the system automatically tracks when a reviewer has taken action on a document, ensuring clear visibility into the review progress.

3.2 Step-by-Step Flow

1. Viewing Comments: When a user opens a document, the system automatically retrieves all existing comments and replies associated with that specific document and displays them in the right-hand panel.
2. Adding Feedback: A user can create a new comment or reply to an existing one. The system saves this input, assigns it an "open" status, and updates the view in real-time.
3. Managing Status: As discussions progress, users can update the status of a comment (e.g., marking an "open" issue as "resolved") to help the team track what still requires attention.
4. Tracking Reviewer Activity: Once a reviewer interacts with the document, the system updates their personal status to "acted," providing a clear indicator that they have participated in the review process.

3.3 Key Concepts

- Comment Threading: The ability to group original comments with their subsequent replies, keeping related feedback organized.
- Comment Status: A label assigned to each comment (such as "open" or "resolved") that helps teams manage the lifecycle of feedback.
- Reviewer Activity Tracking: A system-level flag that records whether a specific reviewer has performed an action on the document, helping project leads monitor participation.
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Real-time Updates: The system’s ability to display new comments or status changes immediately without requiring the user to refresh the page.

3.4 Business Value

- Improved Collaboration: By centralizing feedback in a dedicated panel, team members can discuss specific document sections directly, reducing the need for external communication tools.
- Enhanced Accountability: Tracking reviewer activity ensures that project managers can easily identify who has completed their review and who is still pending.
- Streamlined Workflow: The ability to mark comments as "resolved" allows teams to focus on outstanding issues, ensuring that no critical feedback is overlooked during the review cycle.

BUSINESS REQUIREMENTS DOCUMENT

Review Actions

Prepared by Kolosal AI

Version 1.0 **Date** October 13, 2026 **Status** Draft **Classification** Internal

1. COMPONENTS

1.1 Overview

The Review Actions bar is a navigation and utility tool located at the top of the document review page. Its primary purpose is to provide clear context about the document you are currently viewing and to allow you to easily compare different iterations of that document side-by-side.

1.2 Step-by-Step Flow

1. Identify the Document: The bar displays the document name and its unique ID so you always know which file you are working on.
2. View Current Status: You can see the current version number and the document's status (e.g., Draft, Approved) directly in the bar.
3. Access Comparison Tools: By clicking the "Compare Versions" button, a list of all previous versions of the document appears.
4. Select for Comparison: From the list, you can select any historical version. The system will then automatically display that version alongside your current document for a side-by-side review.

1.3 Key Concepts

- Document Metadata: The essential information about a file, including its name, current version number, and status, which helps maintain clarity during the review process.
- Version History: A complete record of all iterations of a document, allowing users to track changes over time.
- Side-by-Side Comparison: A viewing mode that allows you to see two different versions of a document simultaneously to easily identify updates or modifications.
- Status Badge: A visual indicator that shows the current state of a specific document version, helping you distinguish between drafts, final versions, or other stages.

1.4 Business Value

This feature streamlines the document review process by eliminating the need to manually search for or toggle between different files. By providing instant access to version history and a side-by-side comparison tool, it reduces the risk of human error, ensures that reviewers are always looking at the correct iteration, and significantly speeds up the time required to verify changes between document versions.

2. Logic

2.1 Overview

The "Compare Versions" feature in the Review Action Bar allows users to easily navigate and evaluate the history of a document. By providing a clear list of all available versions, users can quickly switch between different iterations to see how a document has evolved over time. This feature ensures that reviewers always have the context of previous changes at their fingertips.

2.2 Step-by-Step Flow

1. Trigger: The user clicks the "Compare Versions" button within the Review Action Bar.
2. Data Retrieval: The system automatically identifies the current document and fetches the complete history of all associated versions.
3. Processing: The system organizes the version data, including the version number, its current status, and whether it is the most recent version.
4. Display: A dropdown menu appears, presenting the list of versions clearly so the user can select one for comparison.
5. Comparison: Upon selecting a different version, the system displays that version side-by-side with the current document for easy review.

2.3 Key Concepts

- Version History: A chronological record of all saved iterations of a document, allowing users to track changes.
- Status Badge: A visual indicator that shows the current state of a specific version (e.g., draft, pending, or approved).
- Current Version Indicator: A label that highlights which version is the most up-to-date, helping users distinguish it from historical drafts.
- Side-by-Side Comparison: A viewing mode that places two versions of a document next to each other to make identifying differences straightforward.

2.4 Business Value

This feature significantly improves the efficiency and accuracy of the document review process. By enabling side-by-side comparisons, it eliminates the need for manual cross-referencing, reduces the risk of human error, and provides reviewers with the necessary transparency to make informed decisions based on the document's evolution.